Abeer Sethia

Bangalore, India — +91-9148019367 — abeersethia3105@gmail.com
github.com/abeersethia — linkedin.com/in/abeer-sethia

PROFILE

B.Tech Data Science Engineering student (MIT Manipal, 2026) with **research internships at IISc Bangalore and IIT Kharagpur**, and a **clinical diagnostics internship at Philips**. Core expertise in deep learning, computer vision, and signal reconstruction, building validated models, deploying inference systems, and translating research into working pipelines. Experienced across PyTorch, distributed systems, and end-to-end ML workflows from prototype to production.

EDUCATION

Manipal Institute of Technology, Bachelors of Technology in Data Science and Engineering — 2022–2026
Minor in Multimodal Intelligent Systems — *A+ Grade Received*

Select Coursework: Natural Language Processing, Computer Vision, Machine Learning, Deep Learning, Mathematical Foundations for Data Science, Biostatistics, Big Data Analytics, Data Structures and Algorithms, Reinforcement Learning

TECHNICAL SKILLS

- **Languages & Frameworks:** Python, PyTorch, PyDICOM, TensorFlow, Java, C++, SQL, PySpark, R, Tableau
- **Distributed Systems:** Hadoop, Apache Spark, Hive, Pig
- **Backend & Databases:** RESTful Flask APIs, Flask web apps, FastAPI, Uvicorn
- **Machine Learning & LLMs:** Computer Vision, Natural Language Processing, Transformers (Hugging Face), GNN
- **DevOps & Tools:** Git, Docker, Weights & Biases, ITK Snap, 3D Slicer, QGIS

PROFESSIONAL EXPERIENCE

Clinical Diagnostics Project Intern, **Philips** — March–July 2025

- Achieved Dice 0.92 (IoU 0.89) on liver lesion segmentation across CT and MRI by fine-tuning nnU-Net and MedSAM2 on 500+ clinical scans with DICOM preprocessing and demographic augmentation, deployed as a real-time Flask inference application for clinical visualization.

RESEARCH EXPERIENCE

Research Intern – AI, UTSAAH Lab, **Indian Institute of Science Bangalore** — December 2025–Present

- Developing a trackerless ultrasound pose estimation model for fetal imaging by combining 3D spatial reconstruction with **self-supervised representation learning** on B-mode sequences, evaluated against Prevost et al.'s pose estimation benchmark across 5 phantom configurations.

Research Intern - AI, KLIV Lab, **Indian Institute of Technology Kharagpur** — April–October 2025

- Designed and implemented a temporal deep learning architecture for nonlinear dynamical system reconstruction using differentiable topology-preserving loss functions, recovering hidden 3D state-space dynamics from a single observed signal with 98.3% correlation across principal dimensions and 93.4% on the third dimension at up to 8:1 latent compression.

PROJECTS

- **DropEdge++ with Johnson–Lindenstrauss Projection (Minor Specialization Project)**: Developed a GNN framework combining topology-aware edge dropout (DropEdge++) with JL-based random feature projection to mitigate oversmoothing and improve computational efficiency. Achieved up to 8% accuracy gains and 3× faster training on the Cora citation network benchmark.
- **HVQ-AP: Hippocampal Volume Quantification for Alzheimer's Progression**: Quantified hippocampal volumetric progression in Alzheimer's patients from 3D MRI by building an end-to-end pipeline covering DICOM ingestion (PyDICOM), 3D CNN segmentation, and longitudinal volume tracking, achieving Dice 91.6% and Jaccard 84.6% on the Medical Segmentation Decathlon benchmark, matching clinical-grade validation standards. **GitHub**
- **Product Review Sentiment Classifier**: Built a zero-shot multi-label sentiment classifier achieving 85–95% topic accuracy across 3 product categories by ensembling BART, DeBERTa, and DistilBERT NLI models with a PyTorch inference backend, deployed via RESTful Flask API and Streamlit. **GitHub**

CO/EXTRA CURRICULAR

- **Expertise Head – Research Society Manipal:** Mentored 20+ juniors in research methodology, scientific writing, and critical analysis, contributing to improved research quality and output.
- **Data Analyst – IAESTE LC Manipal:** Automated IAESTE LC Manipal's outreach operations across 1,000+ members by building Google Apps Script workflows and real-time dashboards replacing manual tracking, increasing outreach efficiency by 30%, cutting response time by 20